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Que 1 `Console.log('A');`
`SetTimeout(CC) => Console.log('B'), 0);`
`Promise.resolve(C).then() => Console.log('C');`
`Console.log('D');`

Output: A

D

C

B

Ans 2. Calc.js $\rightarrow$ function add(a, b) {
 return a+b;

}

function subtract(a, b) {

 return a-b;

}

module.exports = {

 add,

 subtract

};

app.js

const { add, subtract } = require('./Calc');
 console.log(add(4, 5));

Output will be $\rightarrow 9$.

Qno 3. `const express = require('express');`

`const app = express();`

`app.use(express.json());`

`function requireAuth (req, res, next) {
 if (req.headers['x-api-key'] !== 'Secret123') {
 return res.status(401).json({ error: 'Unauthorized' });
 }
 next();
}`

`}`

`app.get('/api/books', (req, res) => {
 res.json({ message: 'Public' });
});`

`}`

`app.post('/api/books', requireAuth, (req, res) => {
 res.json({ message: 'Book Created' });
});`

`}`

`app.listen(5000);`

Qno 4 => `const express = require('express');`

`const app = express();`

`app.use(express.json());`

`let books = [];`

`app.get('/api/books', (req, res) => {
 res.json(books);
});`

`}`

`app.post('/api/books', (req, res) => {`

`const book = {`

`id: book.length + 1,`

`title: req.body.title,`

`author: req.body.author,`

`};
 books.push(book);`

`});`

```

res.status(201).json(book);
});
app.delete('/api/books/:id', (req, res) => {
  const index = books.findIndex(b => b.id === req.params.id);
  if (index === -1)
    return res.status(404).json({ error: 'Book not found' });
  books.splice(index, 1);
  res.json({ message: 'Book deleted' });
});
app.listen(5000);

```

Ques - If the book is not found, book is undefined so book.title causes an error

Correct

```

Get CC -> app.get('/api/books/:id', (req, res) => {
  const book = books.find(b => b.id === req.params.id);
  if (!book) {
    return res.status(404).json({
      message: 'Book not found'
    });
  }
  res.status(200).json(book.title);
});

```

Qno 6 Import React, {useEffect, useState} from 'react';
function BookList() {

const [books, setBooks] = useState([]);
const [loading, setLoading] = useState(false);
const [error, setError] = useState('');

useEffect(() => {

fetch('http://localhost:5000/api/books')

• then (response => {

if (!response.ok) {

throw new Error('Failed to fetch books');

}

return response.json();

});

Qno 7. CORS occurs because React (localhost:3000) and Express (localhost:5000) are different origins.

⇒ Install CORS. → npm install cors.

Express Code → const cors = require('cors');
app.use(cors);

This allows the react to access the Express Api